

GRADE 10 · ICT · MODULE 3.2

# Number Systems & Conversions

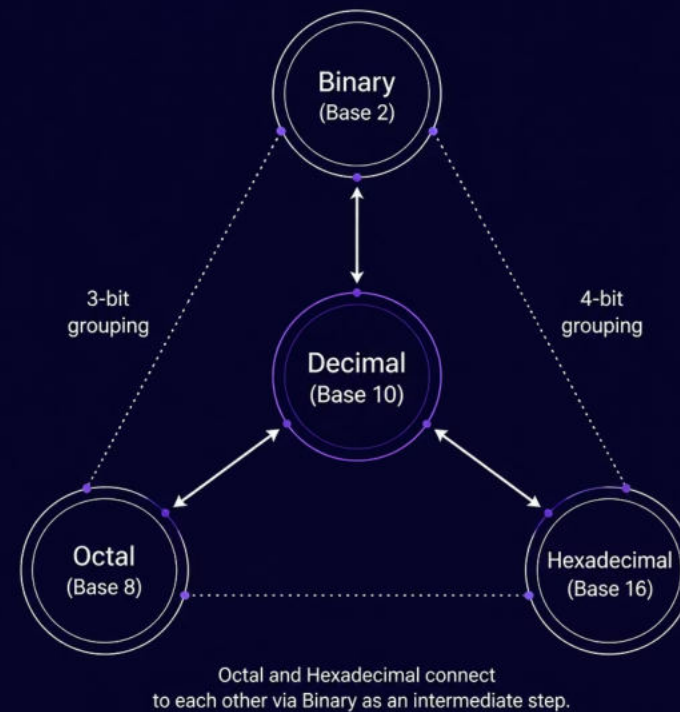
Master conversions between **Decimal**, **Binary**, **Octal**, and **Hexadecimal** using division, multiplication, and bit-grouping methods.

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## OVERVIEW

# The Conversion Map

All four number systems connect through a central hub. **Decimal** uses the **division method** to convert out and **positional multiplication** to convert in. **Binary** ↔ **Octal** groups **3 bits**; **Binary** ↔ **Hex** groups **4 bits**. Octal and Hex convert via Binary as a bridge.



# Decimal to Binary, Octal & Hexadecimal

**Divide by the target base** until the quotient is 0. Read remainders **bottom to top**. For Hex, remainders 10–15 become A–F.

$46_{10} \rightarrow \text{Binary } (\div 2)$

```
2 | 46 → r 0
2 | 23 → r 1
2 | 11 → r 1
2 | 5  → r 1
2 | 2  → r 0
2 | 1  → r 1
4610 = 1011102
```

$158_{10} \rightarrow \text{Octal } (\div 8)$

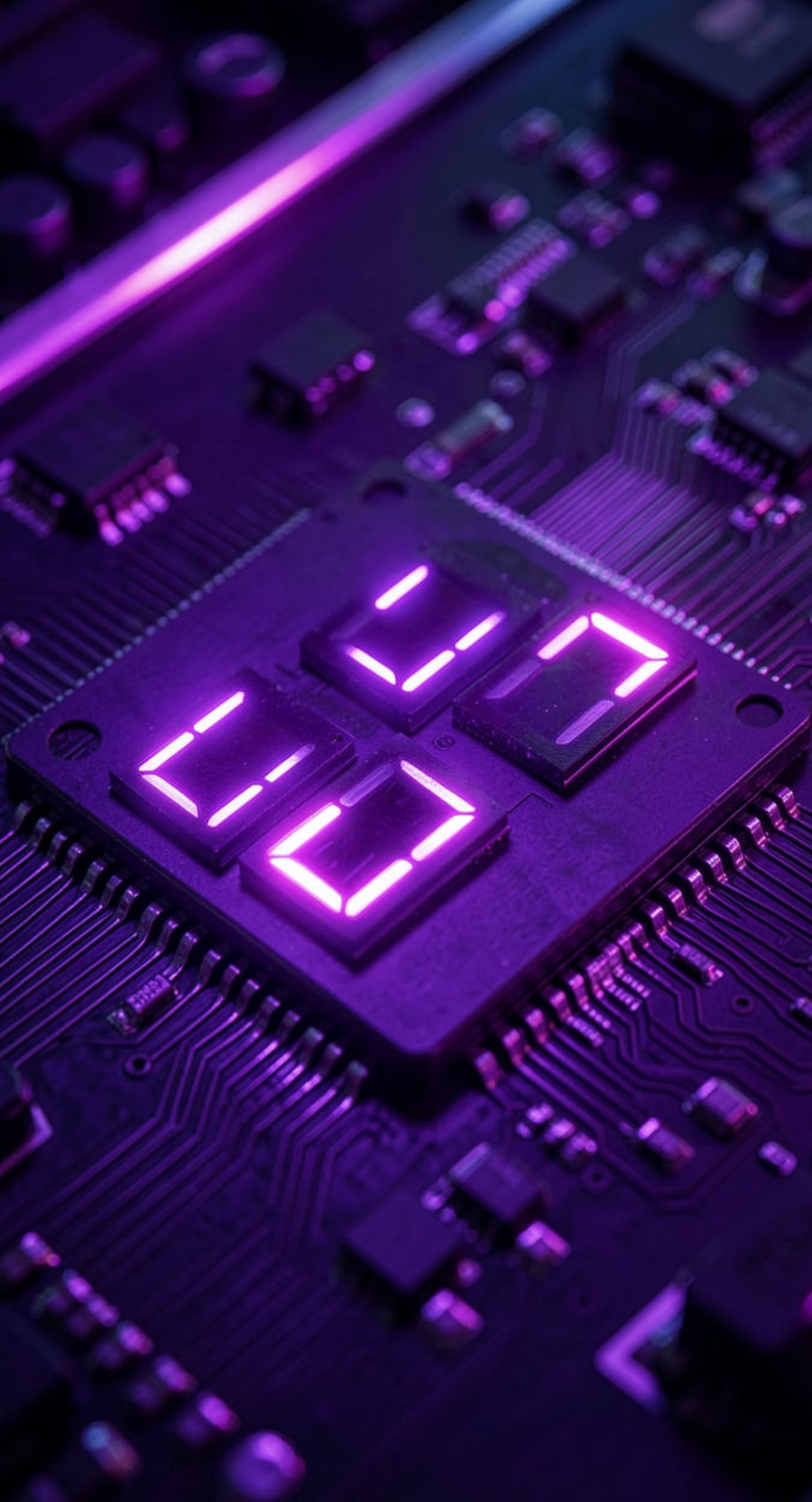
```
8 | 158 → r 6
8 | 19  → r 3
8 | 2   → r 2
15810 = 2368
```

$47_{10} \rightarrow \text{Hex } (\div 16)$

```
16 | 47 → r 15 (F)
16 | 2  → r 2
4710 = 2F16
```



**Try:**  $155_{10} \rightarrow \text{binary}$  ·  $75_{10} \rightarrow \text{octal}$  ·  $256_{10} \rightarrow \text{hex}$



BINARY → DECIMAL

# Binary to Decimal

Multiply each bit by its **positional value** ( $2^n$ , right to left from  $2^0$ ) and sum the results.

Example:  $1101_2$

|        |       |       |       |       |
|--------|-------|-------|-------|-------|
| Bit:   | 1     | 1     | 0     | 1     |
| Power: | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
| Value: | 8     | 4     | 2     | 1     |

$$(1 \times 8) + (1 \times 4) + (0 \times 2) + (1 \times 1) \\ = 8 + 4 + 0 + 1 = 13_{10}$$

## Why It Works

Each position is a power of 2. Add only positions where the bit is **1**.



**Try:**  $101_2 \cdot 111010110_2 \cdot 1010010111_2$

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# Octal & Hexadecimal to Decimal

Multiply each digit by its positional value (**base<sup>n</sup>**) and sum. For Hex, A=10 through F=15.

1275<sub>8</sub> → Decimal

**Digit:** 1   2   7   5  
**Power:** 8<sup>3</sup> 8<sup>2</sup> 8<sup>1</sup> 8<sup>0</sup>  
**Value:** 512   64   8   1

$$(1 \times 512) + (2 \times 64) + (7 \times 8) + (5 \times 1) \\ = 512 + 128 + 56 + 5 = 701_{10}$$

AB2<sub>16</sub> → Decimal

**Digit:** A   B   2  
**Power:** 16<sup>2</sup> 16<sup>1</sup> 16<sup>0</sup>  
**Value:** 256   16   1

$$(10 \times 256) + (11 \times 16) + (2 \times 1) \\ = 2560 + 176 + 2 = 2738_{10}$$

📌 **Try:** 230<sub>8</sub> · 745<sub>8</sub> · 1A<sub>16</sub> · 7EF<sub>16</sub>

# Binary ↔ Octal: Group by 3 Bits

Since  $8 = 2^3$ , each octal digit maps to exactly **3 binary bits**.

## Binary → Octal

Group into 3-bit blocks from the **right**, pad with leading zeros, then convert each group.

101110<sub>2</sub> → 101 110

5 6

Result: 56<sub>8</sub>

## Octal → Binary

Replace each octal digit with its **3-bit** binary equivalent.

457<sub>8</sub> → 4 5 7

100 101 111

Result: 100101111<sub>2</sub>

| Octal | Binary | Octal | Binary |
|-------|--------|-------|--------|
| 0     | 000    | 4     | 100    |
| 1     | 001    | 5     | 101    |
| 2     | 010    | 6     | 110    |
| 3     | 011    | 7     | 111    |

# Binary ↔ Hexadecimal: Group by 4 Bits

Since  $16 = 2^4$ , each hex digit maps to exactly **4 binary bits**.

## Binary → Hex

Group into 4-bit blocks from the **right**, pad with leading zeros, then convert each group.

10110<sub>2</sub> → 0001 0110

1 6

Result: 16<sub>16</sub>

## Hex → Binary

Replace each hex digit with its **4-bit** binary equivalent.

2AE<sub>16</sub> → 2 A E

0010 1010 1110

Result: 1010101110<sub>2</sub>

| Hex | Binary    | Hex | Binary    |
|-----|-----------|-----|-----------|
| 0–7 | 0000–0111 | 8–F | 1000–1111 |
| A   | 1010      | D   | 1101      |
| B   | 1011      | E   | 1110      |
| C   | 1100      | F   | 1111      |

OCTAL ↔ HEXADECIMAL

# Octal ↔ Hexadecimal via Binary

Use **Binary as a bridge**: convert to binary first, then regroup into the target base.

Octal → Hex:  $1057_8$

Step 1 (Octal → Binary):

1 0 5 7  
001 000 101 111

Step 2 (Regroup 4 bits):

0010 0010 1111  
2 2 F

Result:  $1057_8 = 22F_{16}$

Hex → Octal:  $23A_{16}$

Step 1 (Hex → Binary):

2 3 A  
0010 0011 1010

Step 2 (Regroup 3 bits):

001 000 111 010  
1 0 7 2

Result:  $23A_{16} = 1072_8$



**Key Rule:** Octal → Binary uses **3 bits**/digit. Hex → Binary uses **4 bits**/digit.



## PRACTICE

# Worked Examples & Practice

**Which systems allow "23y"?** Invalid in Binary (digit 3 not allowed). Valid in Octal, Decimal, and Hex.

$83_{10} \rightarrow \text{Binary}$

$2 \mid 83 \rightarrow r1 \cdot 2 \mid 41 \rightarrow r1 \cdot 2 \mid 20 \rightarrow r0$   
 $2 \mid 10 \rightarrow r0 \cdot 2 \mid 5 \rightarrow r1 \cdot 2 \mid 2 \rightarrow r0 \cdot 2 \mid 1 \rightarrow r1$   
 $83_{10} = 1010011_2$

Hex Colours (Table 3.13)

- Dark Purple: **#871F78**  $\rightarrow \text{RGB}(135,31,120)$
- Light Pink: **#FFB6C1**  $\rightarrow \text{RGB}(255,182,193)$
- Sky Blue: **#3299CC**  $\rightarrow \text{RGB}(50,153,204)$
- Green: **#00FF00**  $\rightarrow \text{RGB}(0,255,0)$
- Yellow: **#FFEE00**  $\rightarrow \text{RGB}(255,238,0)$



**Try:**  $45_{10} \rightarrow \text{binary, octal, hex}$  ·  $57_8 \rightarrow \text{binary, decimal}$  ·  $2F_{16} \rightarrow \text{binary, decimal}$

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## SUMMARY

# Module 3.2: Key Takeaways

| From        | To          | Method                                    |
|-------------|-------------|-------------------------------------------|
| Decimal     | Any base    | Divide by base; read remainders bottom-up |
| Any base    | Decimal     | Multiply digits by $\text{base}^n$ ; sum  |
| Binary      | Octal / Hex | Group 3 bits / 4 bits                     |
| Octal / Hex | Binary      | Each digit $\rightarrow$ 3 / 4 bits       |
| Octal ↔ Hex | —           | Via Binary bridge                         |

## 3-Bit Rule

Binary ↔ Octal:  $8 = 2^3$

## 4-Bit Rule

Binary ↔ Hex:  $16 = 2^4$

## Bridge Rule

Octal ↔ Hex via Binary